

Problem Statement

Most individuals struggle with **financial planning and budgeting** because their financial data is:

- **Fragmented** across bank apps, wallets, and statements
- Difficult to **analyze manually**
- Lacking **intelligent insights**

As a result, users:

- Cannot track spending effectively
- Fail to maintain budgets
- Miss opportunities to save

Goal: Build an AI-powered platform that **automatically tracks, categorizes, and analyzes financial data**, providing actionable insights

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MEMBER 4

DAY1

Data Ingestion (Plaid API / CSV Upload)



Transaction Storage (Database)



Categorization Engine (LLM)



LangChain ReAct Agent (AI reasoning & tool use)



Streamlit UI (dashboard + insights)

TASK1- `pip install scikit-learn pandas openai langchain`
(VERIFIED WITH NO ERRORS)

TASK 2- Write agent/tools.py

Write 4 function stubs with docstrings only: `get_transactions(user_id, start_date, end_date)`, `get_budget_status(user_id, month)`, `calculate_savings_projection(monthly_savings, months, annual_rate)`, `categorize_spending(description)`. Add a comment above each explaining what it will do on Day 3.

TASK 3-Transaction Categorization Engine

Add "Transaction Categorization Engine" section. Explain: use few-shot LLM prompting to classify raw text. Show 3 examples: "SWIGGY ORDER 4421" → Food, "UBER TRIP 8823" → Transport, "BESCOM BILL" → Housing

Transaction Categorization Engine

The **Transaction Categorization Engine** is responsible for automatically classifying raw financial transaction descriptions into meaningful spending categories, helping users understand their spending patterns without manual effort.

Objective

Financial transaction data often comes in **unstructured text format** (e.g., merchant names, codes, or abbreviations), making it difficult to track expenses or budget effectively. This engine leverages AI to convert raw text into structured categories.

Approach: Few-Shot LLM Prompting

Instead of relying on rigid keyword rules, we use a **few-shot learning approach with a Large Language Model (LLM)**. The model is given a few labeled examples and generalizes to classify unseen transaction descriptions correctly.

Few-Shot Prompt Example

The LLM receives:

Instruction: "Categorize the following transaction into one of these categories: Food, Transport, Housing, Entertainment, Others."

Examples:

- "SWIGGY ORDER 4421" → Food
- "UBER TRIP 8823" → Transport
- "BESCOM BILL" → Housing

New input: "DOMINOS PIZZA 324"

Model output: Food

How It Works

1. **Input:** User transaction (from bank, wallet, or CSV)
2. **Processing:** LLM classifies the description using few-shot examples
3. **Output:** Category label stored in the database and used for analytics

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Member-5 ROLE - Architecture Design Maker.

TASK -1

1. ML CATEGORIZATION- System that can **read transaction names and guess what they are**

→ We will implement a **transaction categorization engine** to automatically classify raw financial transactions into predefined categories such as **Food, Transport, Shopping, Utilities, and Income**.

In the initial phase, we will use a **rule-based keyword matching approach** to ensure fast and reliable performance. For example:

- "UBER TRIP" → Transport
- "SWIGGY ORDER" → Food

This approach is simple, efficient, and suitable for real-time processing during early development.

To enhance scalability and intelligence, the system will later be extended using **LLM-based few-shot prompting via LangChain**, enabling the model to accurately classify previously unseen or ambiguous transaction descriptions.

This hybrid approach ensures both:

- **Accuracy** (through rule-based logic)
- **Flexibility & adaptability** (through LLM integration)

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2. FRONTEND / DASHBOARD

→ **Title:**

"User Interface & Dashboard Design"

The frontend will be developed using Streamlit to provide an interactive and user-friendly dashboard.

Key features include:

- Display of categorized transactions in tabular format
- Visualization of spending patterns using charts
- Category-wise expense breakdown
- Filtering options for better analysis

The dashboard will serve as the primary interface for users to interact with their financial data and gain insights.

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DAY 1:-

STEP1:-MAKE A DATABASE SCHEMA

MAIN TABLE (USERS, TRANSACTIONS, CATEGORIES, BUDGETS)

STEP2:-ER diagram:-

Tumhe dikhana hai:

- **Users → Transactions**
- **Categories → Transactions**
- **Users → Budgets**
- **Categories → Budgets**

Simple relation:

- One user can have many transactions
- One category can have many transactions
- One user can have many budgets
- One category can have many budgets

STEP3:-SQL structure

Is step me tum jo schema banaya hai usko SQL queries me likhogi.

Matlab:

- **CREATE TABLE Users**
- **CREATE TABLE Transactions**
- **CREATE TABLE Categories**
- **CREATE TABLE Budgets**